EPSR-test-ex2 Automatic ereq control with a ceiling (ereqmax)

Program(s): epsr | chained passes: 6 | verification: ALL IDENTICAL (all 6 passes)

What this case tests

Exercises the automatic control of the empirical-potential amplitude `ereq` in the `epsr` engine of EPSR [1,2]: with `ereqstep` > 0 the program steps `ereq` after each run according to the trend of the quality factor in the `.erg` history, biased by `bias`, and bounded by `ereqmin`/`ereqmax` [3]. This is how users let EPSR raise the EP amplitude gradually instead of setting it by hand.

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` lines 2180-2262 (automatic `ereq` step, `ereqmax` ceiling), reading of `ereqmin/ereqmax/ereqstep/bias/sfreq`
- `epsr_standalone_rev3.f`: `lsparabola1` (trend line fitted to the `.erg` history), `ptgnsq` (EP refinement with the new amplitude)
- `read_and_save_files.f`: `read_pcof`, `save_files_others` (writes the new `ereq` to the `.pcof`)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::automatic_ereq`, `Epsr::setup` (control keywords), `Epsr::iterate`
- `potential.rs`: `lsparabola1`, `State::ptgnsq`
- `pcof.rs`: `read_pcof`, `save_pcof_file`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. The EP amplitude starts at `ereq` = 10 kJ/mol and is stepped in units of 1 kJ/mol with the default bias 2 and a ceiling `ereqmax` = 14.

Inputs

Copy of EPSR-test with only the `.pcof` header changed: `ereqmax 14.00` and `ereqstep 1.000 1.000` (both 0 in EPSR-test).

How the accuracy was verified

`commands.txt` (1 program call, repeated 6 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 6 passes.
Six chained passes, each reading the `.pcof` and `.erg` written by the previous one, so every step decision is fed back. The control is active: the EP energy column of the six new `.erg` lines of the Fortran reference reads 10, 10, 11, 11, 11, 12 and the final `.pcof` has `ereq` = 12; the ceiling (14) is therefore checked but not reached in this run. The random numbers drawn by the step rule come from the same `ran1` stream, and all 27 output files are identical after every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.